

Sriram Boddeda

Backend & Distributed Systems Engineer | Norman, Oklahoma
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EDUCATION

University of Oklahoma

Master of Science in Computer Science (GPA: 3.78 / 4.0)

Jan 2023 – Dec 2024

Norman, Oklahoma

EXPERIENCE

Research Assistant — Backend & Distributed Systems

University of Oklahoma

Feb 2025 – Present

Norman, Oklahoma

- Architected **AtlasCollect**, a microservices-based data platform scaling to **1.8k req/sec** by decoupling core processing, file services, and schedulers to maximize system fault tolerance.
- Developed a resource-aware job scheduler in Node.js that pre-calculates memory overhead relative to dataset dimensions, **eliminating 100% of system crashes** during high-load data ingestion.
- Optimized computational throughput by designing an **asynchronous parallel worker system** using Node.js child processes, reducing compute-heavy workflow runtimes by **60%**.
- Built a standalone **CLI/TUI Migration Tool** with integrated file deduplication and secure export/import services, improving storage efficiency by **30%** across distributed environments.
- Implemented real-time task tracking via **Server-Sent Events (SSE)**, providing sub-second progress updates and enhancing system transparency for 500+ active researchers.
- Achieved a **45% reduction in API latency** by implementing a **Redis caching layer** and optimizing PostgreSQL indexing for high-concurrency metadata lookups.

Graduate Research Assistant — Full-Stack Development

University of Oklahoma

Oct 2023 – Dec 2024

Norman, Oklahoma

- Engineered the lab's core web platform using **Next.js** and **Strapi**, implementing Redis caching for performance and reducing manual content updates by **50%** through headless CMS automation.
- Designed reusable conference platforms with **year-based routing and automatic archives**, cutting setup time for recurring events by **60%** while ensuring seamless content migration.
- Automated large-scale outreach and database synchronization using **Selenium and Python**, reducing manual coordination effort by **70%** across multiple workshop cycles.

Software Engineer

Persistent Systems

Dec 2020 – Nov 2022

Pune, India

- Developed backend services for an enterprise **Status Board**; delivered **GraphQL/REST APIs** and **LDAP authentication**, reducing manual infrastructure checks by **70%**.
- Architected enterprise **ECM migration pipelines** using ETL/ELT logic, ensuring **100% data consistency** for millions of records through memory-optimized version synchronization.
- Developed PowerShell scripts to validate migration integrity via automated audits, **reducing QA effort by 60%** and saving **30+ engineering hours** per testing cycle.
- Developed a **.NET Core test-data generator** using **RabbitMQ** to interface with legacy ECM libraries, enabling automated data creation and migration testing through a decoupled messaging layer.
- Implemented incremental migration logic via file version synchronization to detect and migrate updated files, significantly reducing bandwidth overhead across environments.

PROJECTS

AtlasCollect Platform | Node.js, Postgres, Redis, Docker

Live

- Built centralized file services and maintenance workers for dataset analysis; improved transparency for 200+ researchers via real-time Task Trackers.

DApp Marketplace | Solidity, Ethereum, ethers.js, Node.js

GitHub

- Led development of secure P2P trading smart contracts on Ethereum; utilized wallet authentication to secure transactions with zero reported breaches.

SKILLS

Languages: Python, Java, JavaScript, TypeScript, C#, SQL (PostgreSQL/MySQL)

Backend & Design: Node.js, .NET Core, **System Design (Scalability/Fault Tolerance)**, REST/gRPC/GraphQL

Distributed Systems: Microservices, Event-Driven Architecture, Kafka, RabbitMQ, Redis (Caching)

Infrastructure: Docker, Kubernetes, AWS, GitHub Actions (CI/CD), Linux CLI

Testing & Security: Unit/Integration Testing (Jest/PyTest), RBAC, OAuth2/JWT

Core Strengths: Data Structures & Algorithms, OOP, Performance Optimization, Database Indexing